Liver Disease Prediction Web Application

Comprehensive Project Report & Technical Documentation

Project Domain: Healthcare / Clinical Decision Support	**Date:** October 2026
Backend: Python 3.14, Flask, Scikit-Learn, Pandas	**Frontend:** HTML5, CSS3, JavaScript (ES6+)
Dataset: Indian Liver Patient Dataset (UCI ILPD)	**Selected Model:** Logistic Regression (83.0% ROC-AUC)

Important Ethical & Clinical Disclaimer: *This web application and machine learning model are developed strictly as an educational and academic demonstration. It is NOT a certified medical diagnostic system and must NEVER replace clinical diagnosis, laboratory pathology, or medical judgment by a licensed physician.*

1. Executive Summary & Objectives

Liver disease represents a major global health challenge, often remaining asymptomatic until advanced stages. The objective of this project is to develop an accessible, interactive, and clinically grounded predictive system capable of analyzing routine liver function test (LFT) biomarkers to provide statistical risk stratification.

The system combines a rigorous Scikit-Learn machine learning pipeline with a responsive Flask web application. It features live physiological input validation, instant biomarker deviation assessment against standard medical reference intervals, and clear classification output as **'Liver Disease Predicted'** or **'No Liver Disease Predicted'**.

2. Dataset & Clinical Biomarkers

The model was trained on the benchmark **Indian Liver Patient Dataset (ILPD)** sourced from the UCI Machine Learning Repository. The dataset comprises 583 patient records (416 liver disease cases and 167 non-disease cases) encompassing 10 vital demographic and laboratory parameters:

Biomarker	Unit	Normal Range	Clinical Significance
Age	Years	1 – 115	Demographic baseline factor
Gender	Binary	Male / Female	Male=1, Female=0 encoding
Total Bilirubin (TB)	mg/dL	0.1 – 1.2	Hemoglobin breakdown product; elevated in jaundice
Direct Bilirubin (DB)	mg/dL	0.0 – 0.3	Conjugated bilirubin; reflects biliary excretion
ALP (Alkaline Phosphatase)	IU/L	44 – 147	Bile duct enzyme; elevated in cholestasis
ALT / SGPT	IU/L	7 – 56	Hepatocellular necrosis & injury marker
AST / SGOT	IU/L	10 – 40	Intracellular enzyme; elevated in liver injury
Total Proteins (TP)	g/dL	6.0 – 8.3	Total serum albumin + globulins
Albumin (ALB)	g/dL	3.5 – 5.0	Synthesized by liver; indicates synthetic function
A/G Ratio	Ratio	1.0 – 2.5	Albumin/Globulin; inverted (<1.0) in cirrhosis

3. Machine Learning Pipeline & Actual Evaluation Results

A rigorous empirical workflow was executed with zero fabricated scores. Missing values in the A/G ratio were addressed using median imputation (median = 0.93). Numerical predictors were normalized using **StandardScaler**. A stratified 80/20 train/test split was established with 5-fold Stratified Cross-Validation.

Model Classifier	5-Fold CV AUC	Test Accuracy	Precision	Recall	F1-Score	Test ROC-AUC
Logistic Regression (Selected)	0.7133	**73.50%**	**91.94%**	68.67%	0.7862	**0.8303 (83.0%)**
Random Forest (150 trees)	0.7266	71.79%	86.76%	71.08%	0.7815	0.8122 (81.2%)
Gradient Boosting (100 trees)	0.7101	66.67%	71.15%	89.16%	0.7914	0.7505 (75.1%)

Model Selection Rationale: Logistic Regression with balanced weighting yielded the best ROC-AUC (0.8303) and highest precision (91.94%). In medical decision support, high precision prevents undue alarm by dramatically suppressing false positives.

Key Biomarker Feature Importance (Normalized Coefficients):

Feature	Weight	Clinical Impact
ALT / SGPT Enzyme	1.168	Primary indicator of acute/chronic hepatocellular injury
AST / SGOT Enzyme	0.853	Correlates with parenchymal hepatic necrosis
Direct Bilirubin	0.708	Strong predictor of cholestasis & biliary excretion disorder
Albumin	0.633	Reflects synthetic capacity of the liver parenchyma
Total Proteins	0.521	Serum protein homeostasis indicator

4. System Architecture & Interactive Features

1. Physiological Input Validation: The application prevents invalid lab entries on both frontend and backend. Direct Bilirubin cannot exceed Total Bilirubin; Albumin cannot exceed Total Protein; non-physiological values are rejected.

2. Dynamic Risk Gauge & Biomarker Flags: Inference computes disease probability (0% - 100%). The interface compares all 10 patient biomarkers against reference ranges and generates contextual guidance.

3. 1-Click Presets & Auto A/G Calculation: Clinical evaluators can click 'Normal / Healthy', 'Elevated Enzymes (Disease)', or 'Borderline Case' to populate the form instantly. An 'Auto-compute' shortcut derives A/G ratio from TP and Albumin.

5. Project Directory Structure

```
Liverdease_prediction/
■■■ app.py # Flask Application Server & Prediction API
■■■ train_model.py # ML Pipeline Training & Evaluation Script
■■■ test_app.py # Automated Unit & Integration Tests (7 tests)
■■■ requirements.txt # Package Dependencies (Flask, Scikit-learn, etc.)
■■■ run.bat # Windows 1-Click Server Launcher
■■■ README.md # Technical Documentation & User Guide
■■■ data/
■ ■■■ indian_liver_patient.csv# UCI ILPD Dataset (583 records)
■■■ models/
■ ■■■ liver_model.joblib # Serialized ML Pipeline & Scaler
■ ■■■ metrics.json # Empirical Validation Metrics & Weights
■■■ templates/
■ ■■■ index.html # Responsive HTML5 Clinical Dashboard
■■■ static/
■■■ css/style.css # Clean CSS3 Responsive Styling & Badges
■■■ js/main.js # Client Validation, Presets & AJAX Controller
```

6. Verification & Automated Test Suite

The project includes an automated test suite (test_app.py) verifying endpoints, validation boundaries, and inference correctness. All 7 tests pass with 100% success:

Test Case	Endpoint Tested	Input Tested	Result
test_home_page_loads	GET /	Root URL request	PASSED (200 OK)
test_health_endpoint	GET /api/health	Health check probe	PASSED (200 OK)
test_metrics_endpoint	GET /api/metrics	Metrics JSON probe	PASSED (200 OK)
test_prediction_healthy	POST /predict	Normal biomarker panel	PASSED ('No Disease')
test_prediction_disease	POST /predict	Elevated enzyme panel	PASSED ('Disease')
test_validation_db_exceeds_tb	POST /predict	DB > TB invalid physiology	PASSED (400 Rejected)
test_validation_invalid_age	POST /predict	Negative / out-of-range age	PASSED (400 Rejected)

7. Step-by-Step GitHub Upload Guide

To publish this complete project repository to your GitHub profile, execute the following commands in **PowerShell** or **Command Prompt**:

```
# Step 1: Open project directory
cd c:\Users\Hemadharshini\OneDrive\Desktop\Liverdease_prediction

# Step 2: Initialize Git repository
git init

# Step 3: Stage all project files (dataset, code, models, static, templates)
git add .

# Step 4: Create the first commit
git commit -m "Initial commit: Complete Liver Disease Prediction Web Application"

# Step 5: Set branch name to main
git branch -M main

# Step 6: Link to your newly created GitHub repository
# (Replace YOUR_USERNAME and YOUR_REPO with your actual GitHub repository URL)
git remote add origin https://github.com/YOUR_USERNAME/YOUR_REPO.git

# Step 7: Push all code to GitHub
git push -u origin main
```

8. How to Run Locally on Windows

```
Method A: 1-Click Launcher (Easiest)
Double-click the 'run.bat' file inside the project directory.

Method B: Terminal Command
py -m pip install -r requirements.txt
py app.py

Access Web UI in Browser:
http://127.0.0.1:5000
```

Report Conclusion: The Liver Disease Prediction System successfully bridges machine learning classification with an accessible, intuitive web interface. All metrics reflect genuine cross-validation performance on UCI ILPD data.